



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- SpaceX launch data was collected from the SpaceX REST API regarding the SpaceX Falcon 9 rocket providers. Information about the landing of the first stage through various recovery methods were analyzed thoroughly and used to train various Machine Learning models. This data was collected to predict whether SpaceX will reuse the first stage and whether the first stage will land successfully.
- After a predictive analysis on the data collected by utilizing models like Logistic Regression, Support Vector machines, Decision Tree Classifier and K-nearest neighbors we concluded that the Decision Tree Classifier provided the best accuracy overall. Providing an accuracy of 83% on the test data. Through this model we were able to predict successfully if SpaceX will land the first stage successfully.

Introduction

- Companies are innovating by creating an increasing amount of spaceships to facilitate affordable space travel. Out of the major companies engaging in this effort SpaceX stands out. This is primarily due to the reduced cost of launch where a Falcon 9 rocket launch only costed 62 million US dollars. In comparison, other providers cost upwards of 165 million US dollars each. The reason for this cost reduction is primarily due to the reusing of the first stage. Therefore, if it was possible to predict if the first stage landed successfully, it would help determine the cost of a launch.

Problems Addressed:

- What factors influenced landing of payload of Falcon 9?
- How to predict the landing outcome of the Falcon 9 payloads?

Section 1

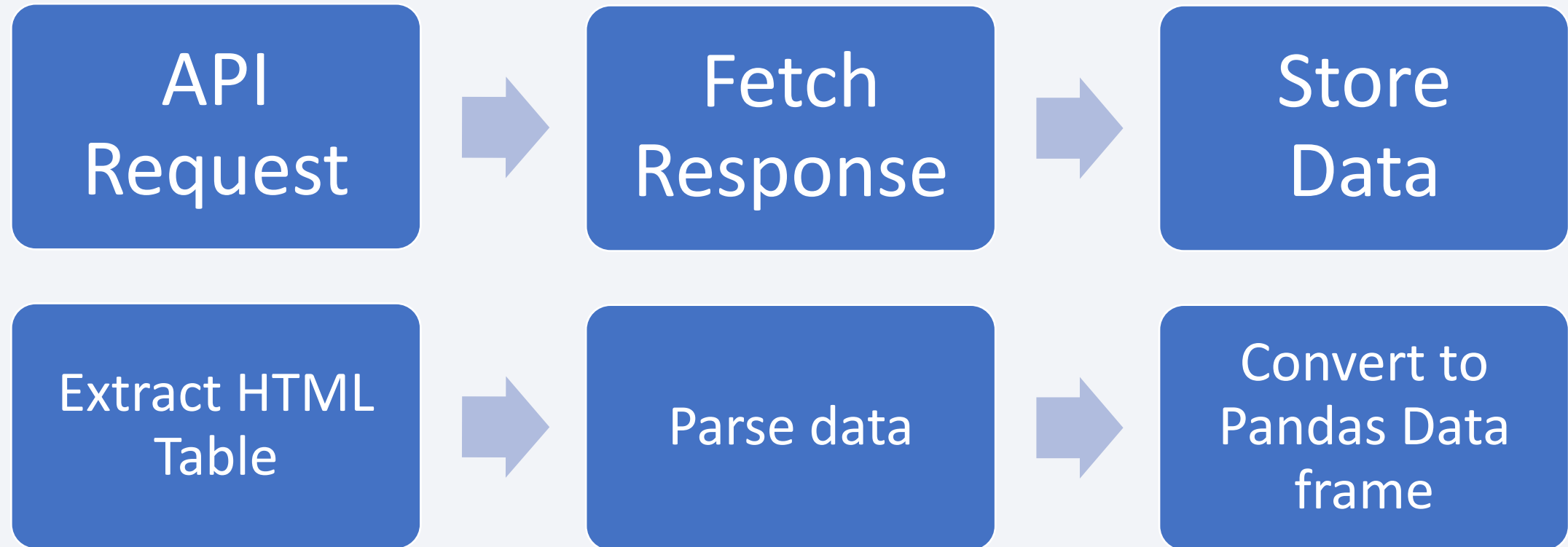
Methodology

Methodology

Executive Summary

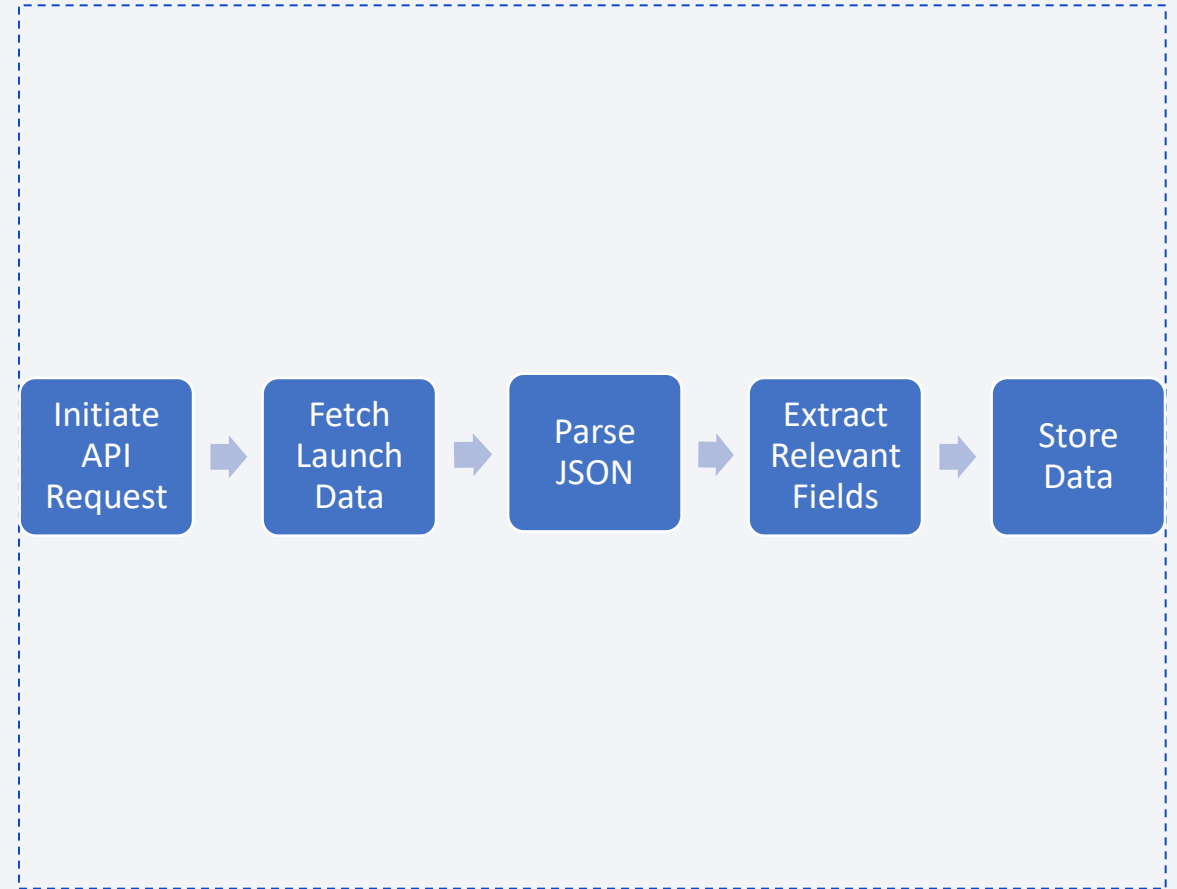
- **Data collection methodology:** Using SpaceX REST API data about launches including information about the rocket used, payload delivered, launch specifications, landing specifications and landing outcomes were collected. Further, information was extracted from wiki pages using web scraping. Then different cases where the booster landed was converted into training labels.
- **Perform exploratory data analysis (EDA) using visualization and SQL:** Visualization on launch success rates, payloads and launch sites were made
- **Perform interactive visual analytics using Folium and Plotly Dash:** Interactive Dashboard was made to view a scatter plot
- **Perform predictive analysis using classification models**
 - Using different machine learning models data was fitted and predictions were made and models were compared.

Data Collection



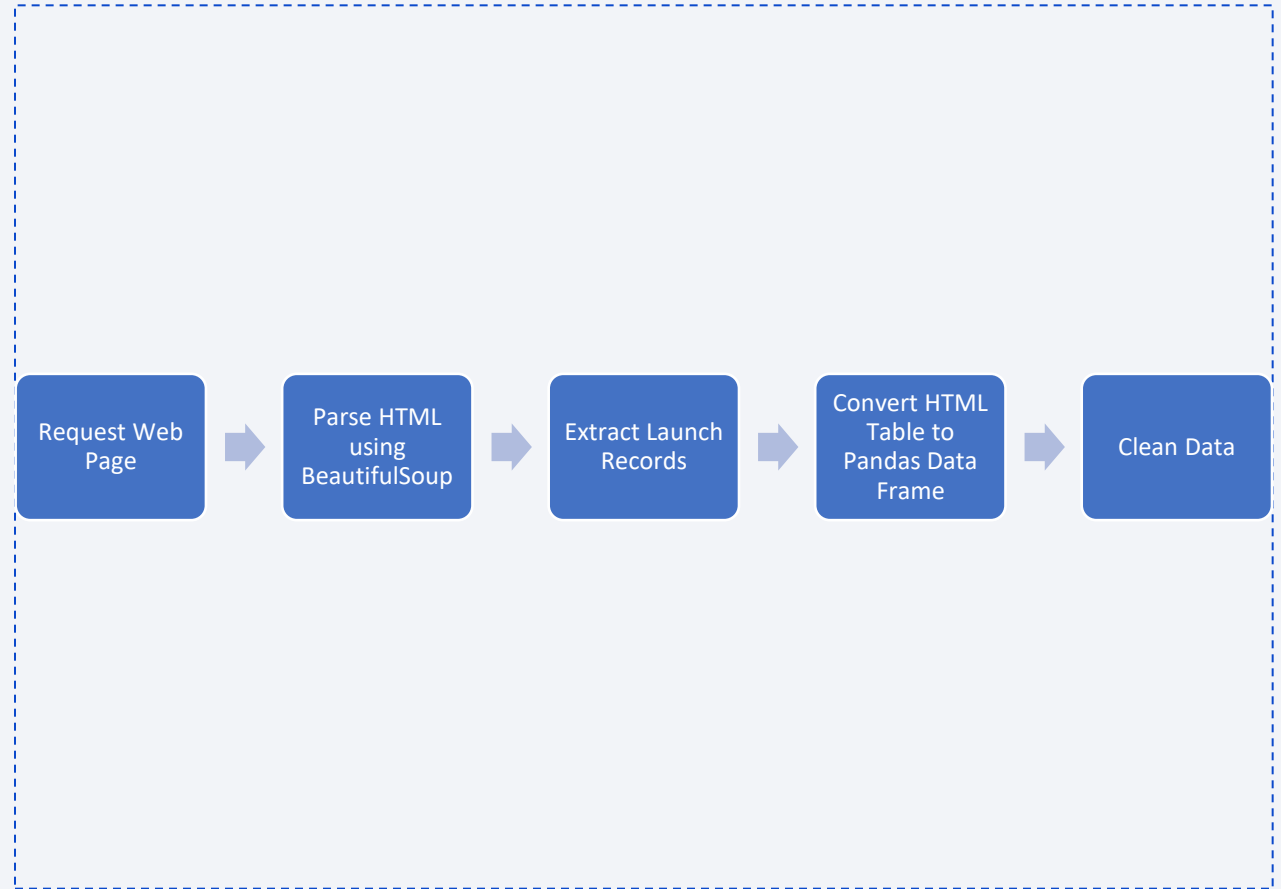
Data Collection – SpaceX API

- GitHub URL:
<https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/jupyter-labs-spacex-data-collection-api.ipynb>



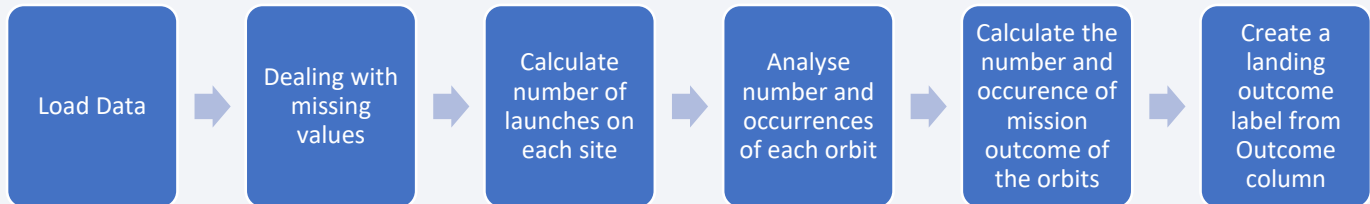
Data Collection - Scraping

- GitHub URL:
<https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/jupyter-labs-webscraping.ipynb>



Data Wrangling

- Data Wrangling process involved the gathering, cleaning and transforming raw unstructured data into a structured reliable format.
- GitHub URL:
<https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

1. Scatter plots: Used to visualize the relationship between Flight Number and Launch Site , the relationship between Payload Mass and Launch Site, the relationship between FlightNumber and Orbit type, the relationship between Payload Mass and Orbit type
2. Bar Chart: Used to visualize the relationship between success rate of each orbit type
3. Line Chart: Used to visualize the launch success yearly trend

GitHub URL: <https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/edadataviz.ipynb>

EDA with SQL

- Initially, the unique names of each launch site used for space missions were extracted.
- Then records of launch sites that begins with the string CCA were extracted.
- Next, the total payload mass carried by the boosters launched by NASA(CRS) was displayed.
- Next the average payload mass carried by booster version F9 v1.1 was displayed.
- Afterwards, the date when the first successful landing outcome in ground pad was achieved was listed.
- Then the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 were listed.

EDA with SQL

- Afterwards, the total number of successful and failure mission outcomes were listed.
- Next, the booster_versions that have carried the maximum payload mass was listed.
- Then the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015 was listed.
- Finally, the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order was ranked.
- GitHub URL: https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/jupyter_labs_eda_sql_coursera_sqlite.ipynb

Build an Interactive Map with Folium

- All launch sites were marked in a map using `folium.Circle` and `folium.Marker` to add a highlighted circle area with a text label on a specific coordinate. Markers were coloured with green for successful launch and red for unsuccessful launch. Then Marker Clusters were created. Afterwards, distanced between a launch site to its proximities were calculated using `MousePosition` and marker was created displaying distance. Then a `PolyLine` was drawn from the launch site to the proximities such as coastline, closest city, railway, highway.
- Folium Circle and Markers were added to identify coordinates in the map. Marker Clusters were used to simplify the map containing many markers having the same coordinates. `PolyLine` was used to visually show the distance between the site and proximities.
- GitHub URL: https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/lab_jupyter_launch_site_location.ipynb

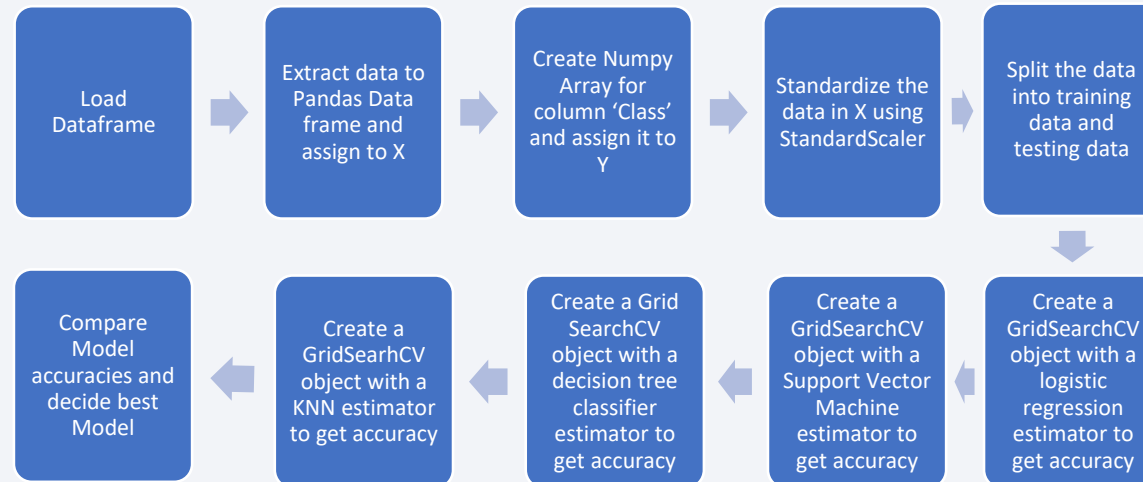
Build a Dashboard with Plotly Dash

- This dashboard application contains input components such as a dropdown list and a range slider to interact with a pie chart and a scatter point chart.
- We have four different launch sites and we would like to first see which one has the largest success count. Then, we would like to select one specific site and check its detailed success rate. This is achieved by the pie chart controlled by the dropdown.
- The Range Slider and scatter point chart are used to be able to easily select different payload range and see if we can identify some visual patterns.
- GitHub URL: <https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/spacex-dash-app.py>

Predictive Analysis (Classification)

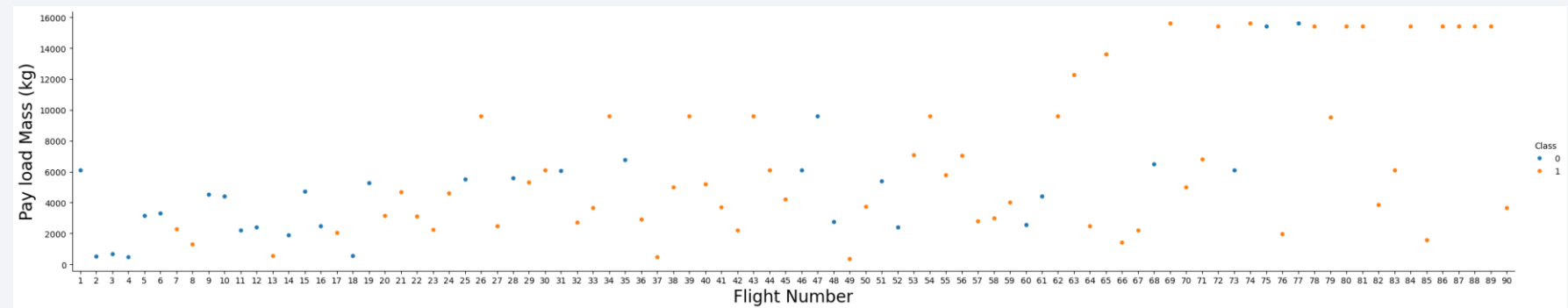
- The data was imported and the X variable was assigned with the pandas dataframe while the Y variable was assigned a numpy array with the class values. Then the data was split into training data and testing data. Afterwards, the data was fitted into a GridSearchCV object (for hyperparameter tuning) with different models such as Logistic Regression, Support Vector Machine, Decision Tree and K-nearest neighbor. Evaluation metrics like accuracy, precision, recall and F1-score of each model was displayed in confusion matrices. Based on the training accuracy and testing accuracy the best model was selected.

Predictive Analysis (Classification)

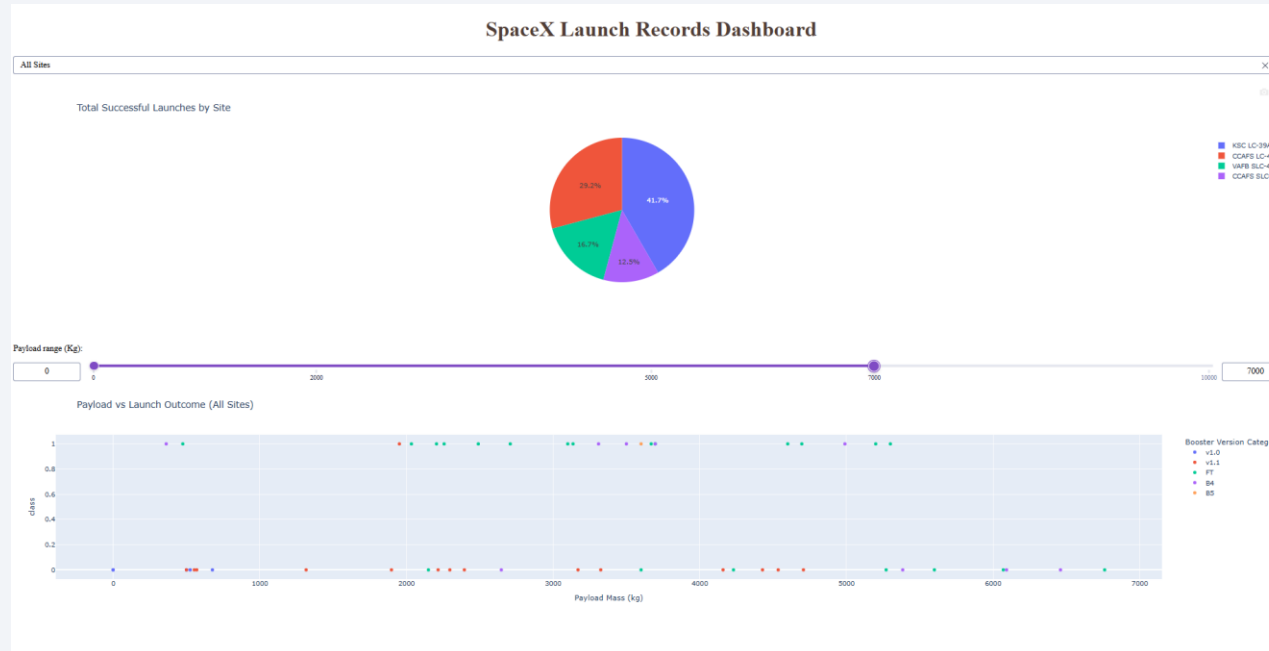


- GitHub URL: https://github.com/Dineth-Wickremasinghe/SpaceX-Analysis/blob/main/Notebooks/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

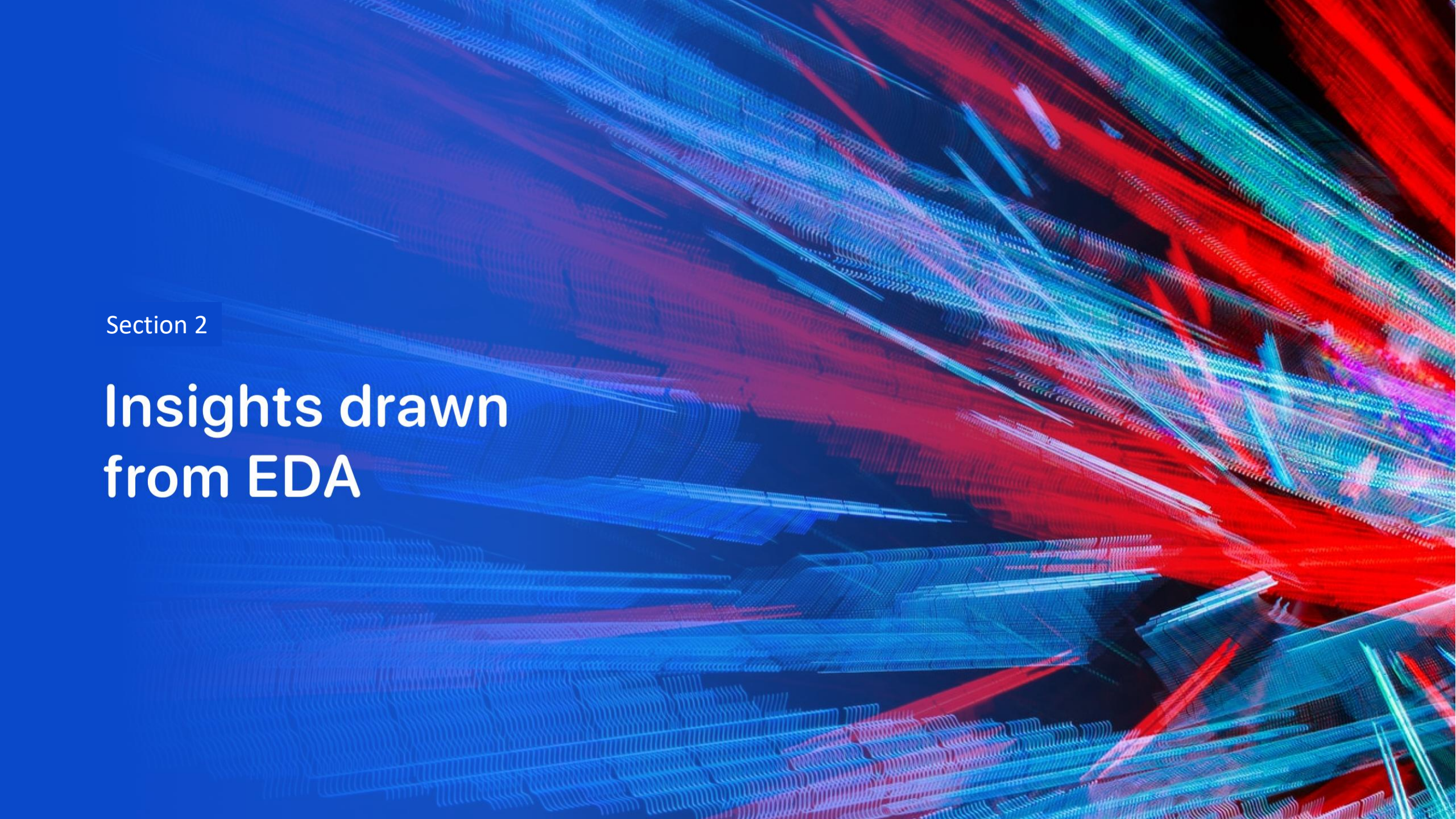
Results



Results



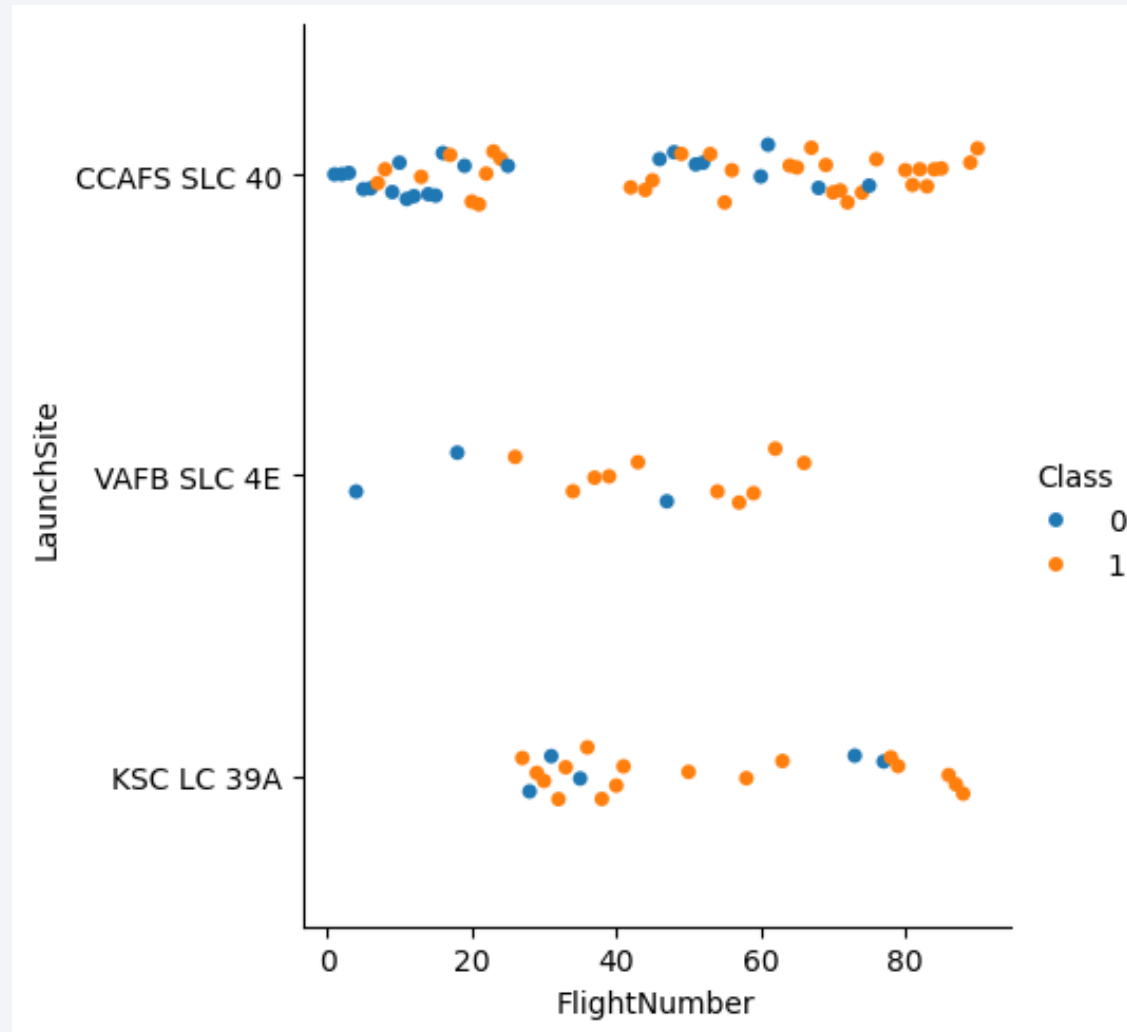
Model with best training and testing accuracy was concluded to be Random Forest Classifier with 0.889 training accuracy and 0.83 testing accuracy.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

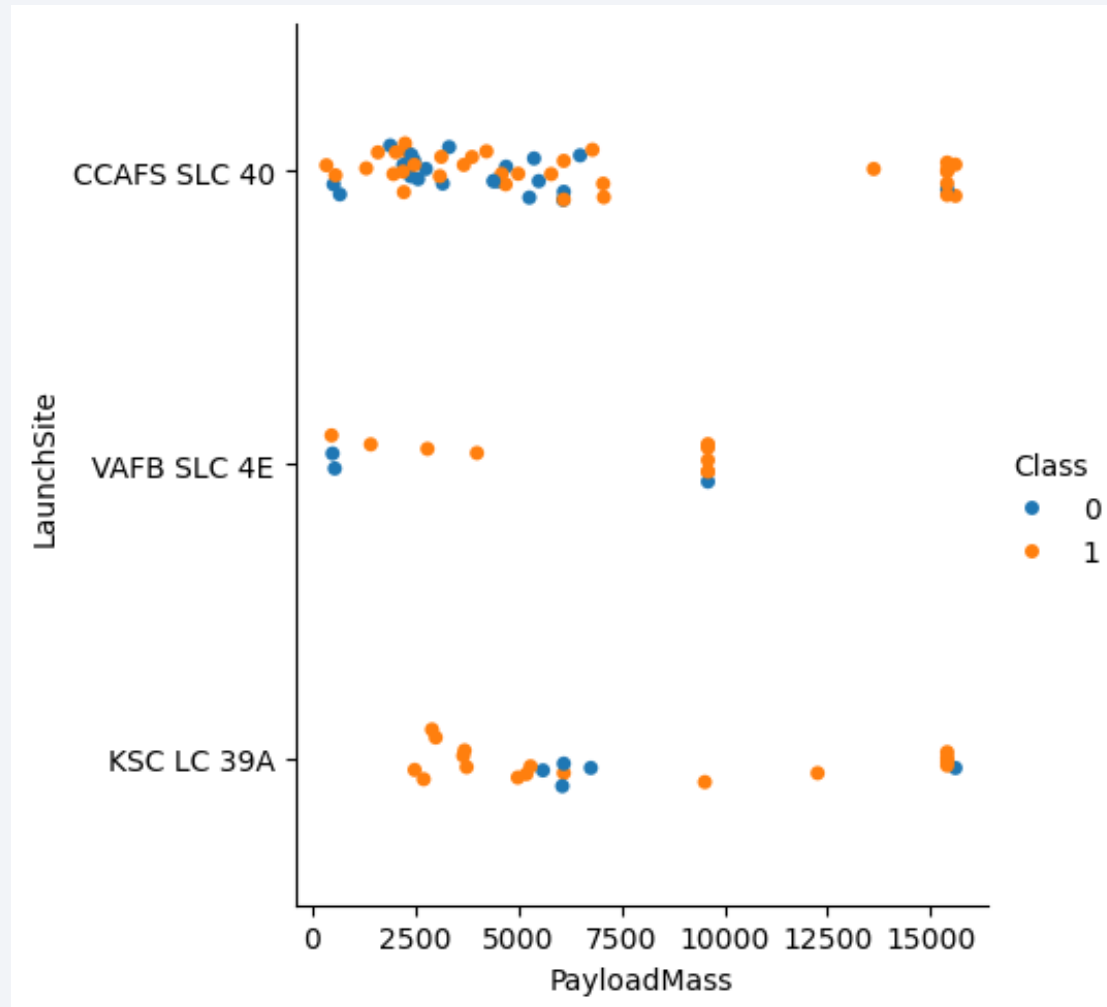
Section 2

Insights drawn from EDA

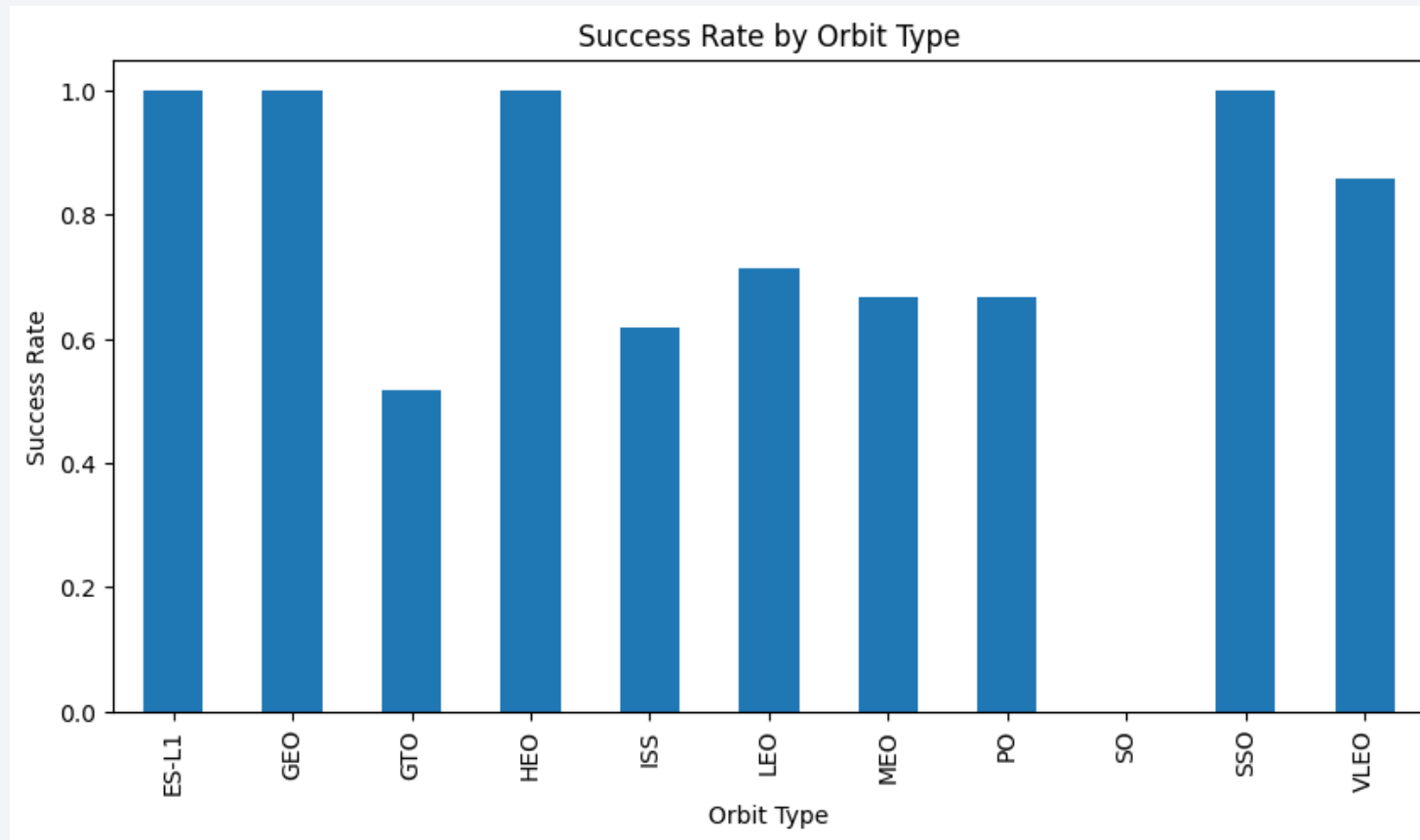
Flight Number vs. Launch Site

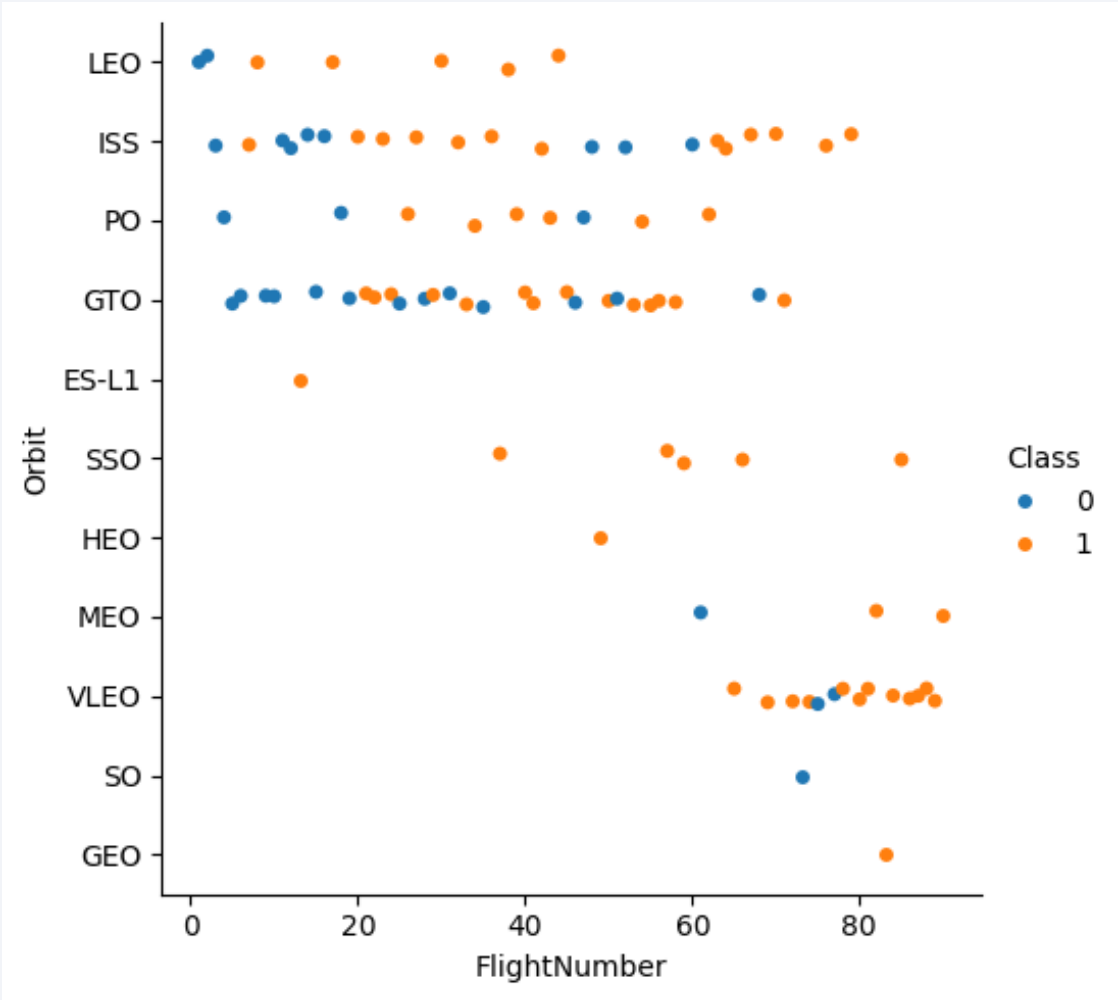


Payload vs. Launch Site

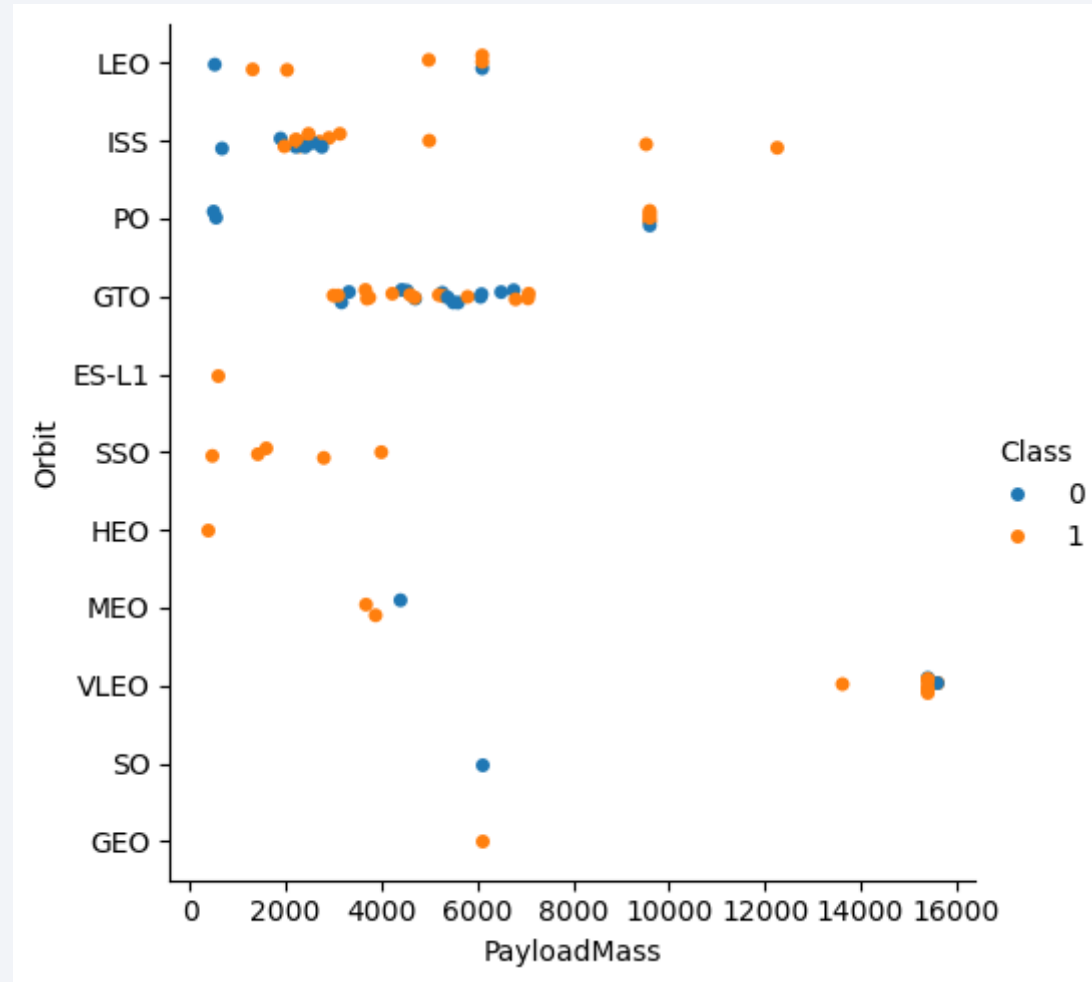


Success Rate vs. Orbit Type

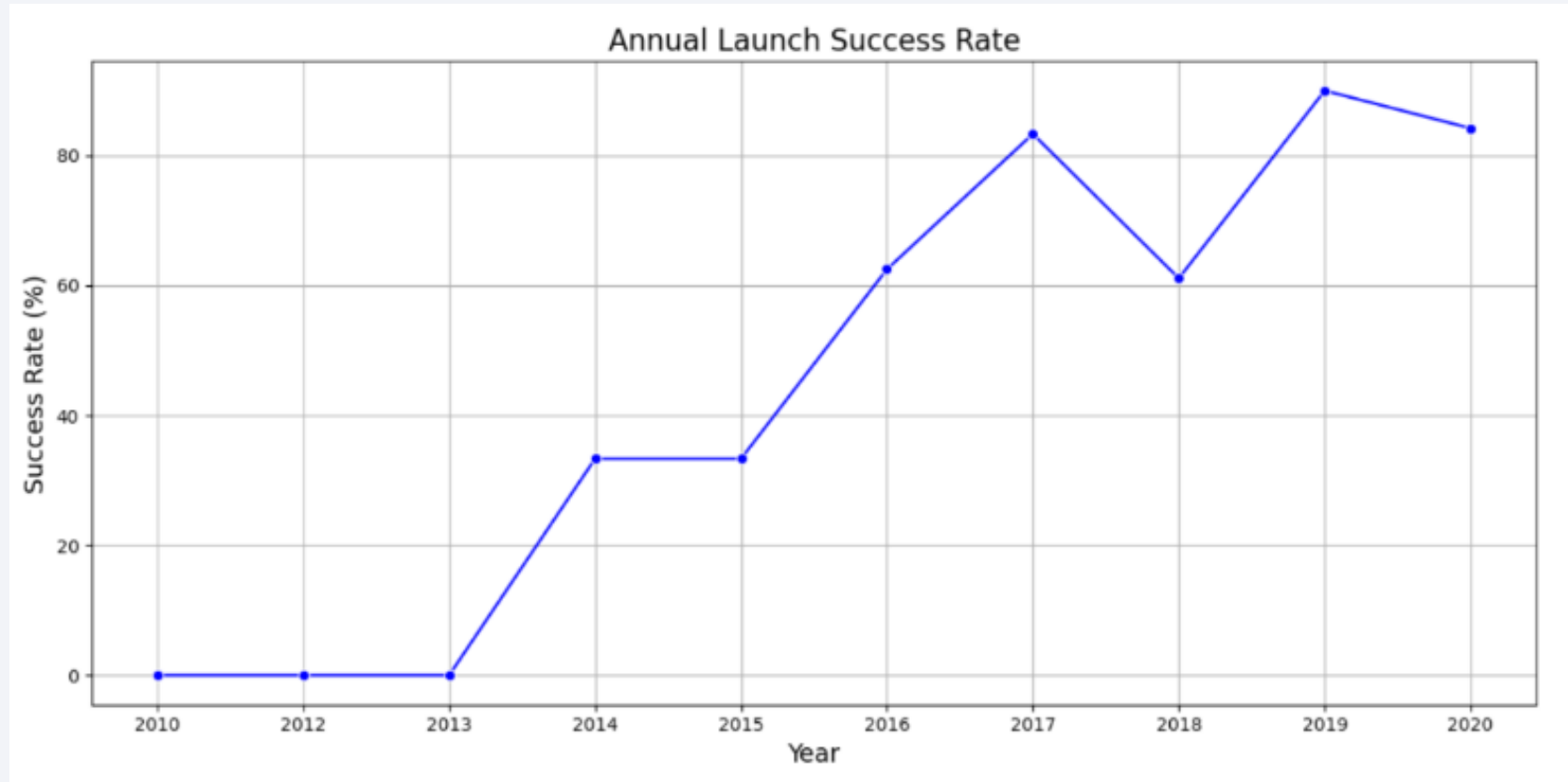




Payload vs. Orbit Type



Launch Success Yearly Trend



All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

Python

* [sqlite:///my_data1.db](#)

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
%sql SELECT SUM("Payload_Mass__kg_") FROM SPACEXTBL WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
SUM("Payload_Mass__kg_")
```

```
45596
```

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG("Payload_Mass__kg_") FROM SPACEXTBL WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

AVG("Payload_Mass__kg_")

2928.4

First Successful Ground Landing Date

```
%sql SELECT MIN("Date") FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
MIN("Date")
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT "Booster_Version" FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (drone ship)' AND "Payload_Mass__kg_" > 4000 AND "Payload_Mass__kg_"
```

Python

```
* sqlite:///my\_data1.db
```

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%sql SELECT "Booster_Version" FROM SPACEXTBL WHERE "Payload_Mass__kg_" = ( SELECT MAX("Payload_Mass__kg_") FROM SPACEXTBL );
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

```
%sql SELECT substr("Date",6,2) AS Month, "Booster_Version", "Launch_Site", "Landing_Outcome" FROM SPACEXTBL WHERE substr("Date",1,4)='2015' AND "Lar
```

Python

```
* sqlite:///my\_data1.db
```

Done.

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT "Landing_Outcome", COUNT(*) AS Count FROM SPACEXTBL WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER
```

Python

```
* sqlite:///my\_data1.db
```

Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

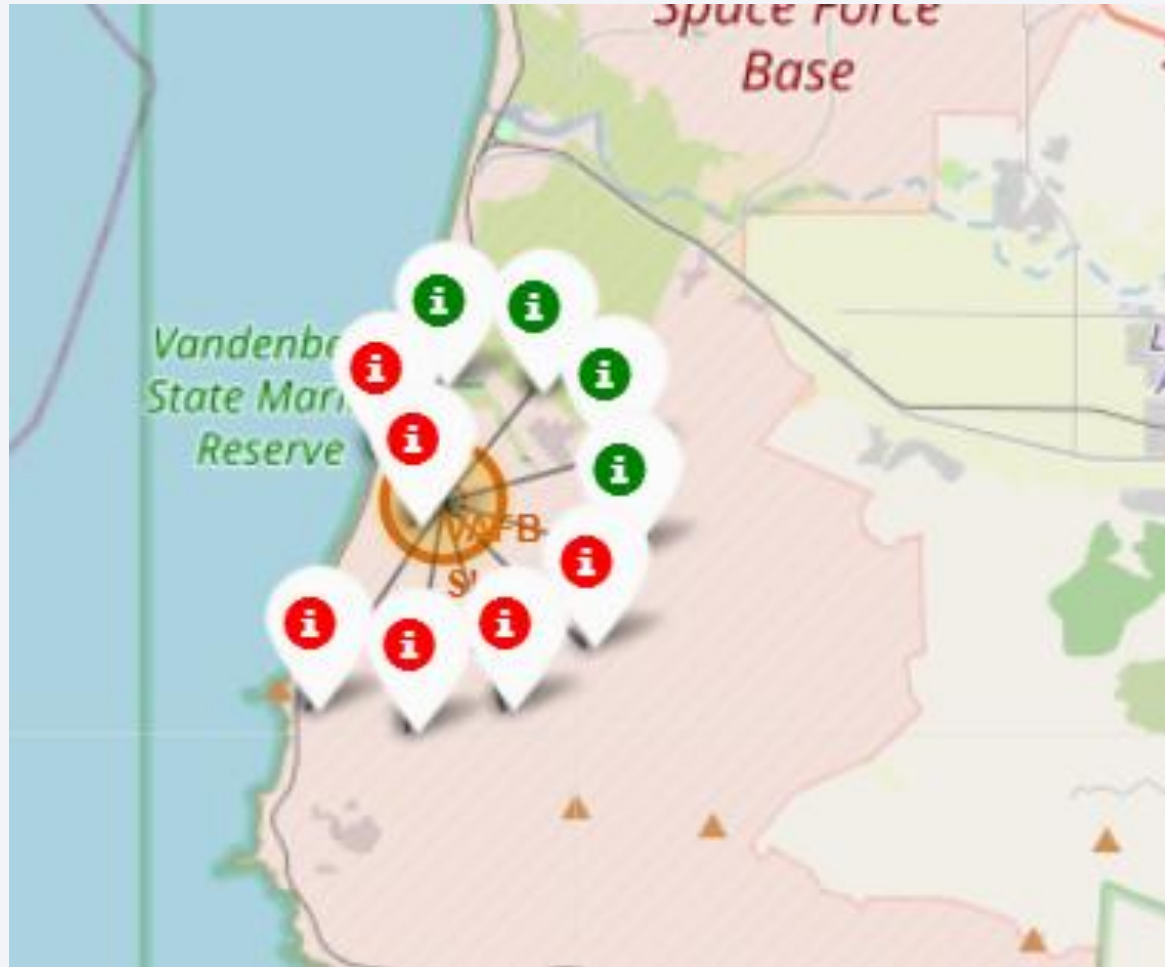
Section 3

Launch Sites Proximities Analysis

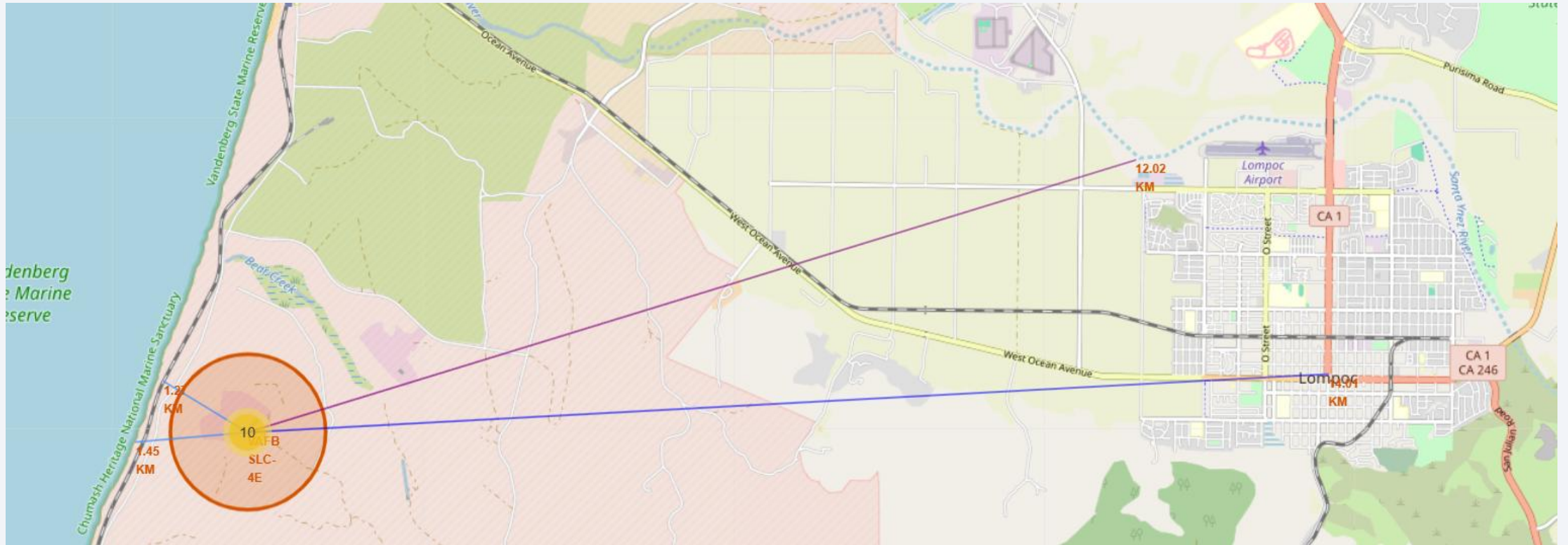
Folium Map Zoomed Out

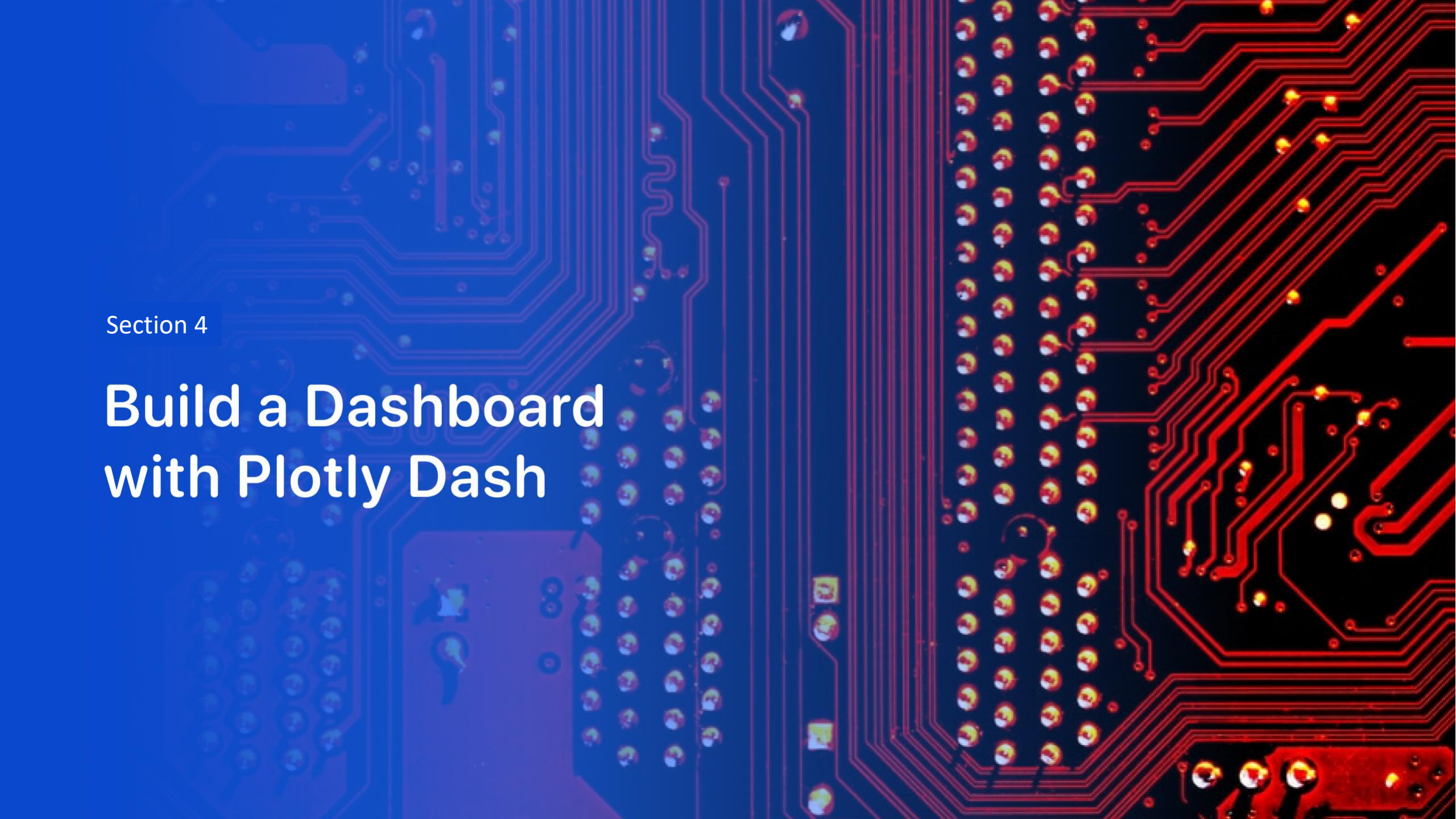


Folium Map Zoomed in



Folium Map showing distances between sites and proximities

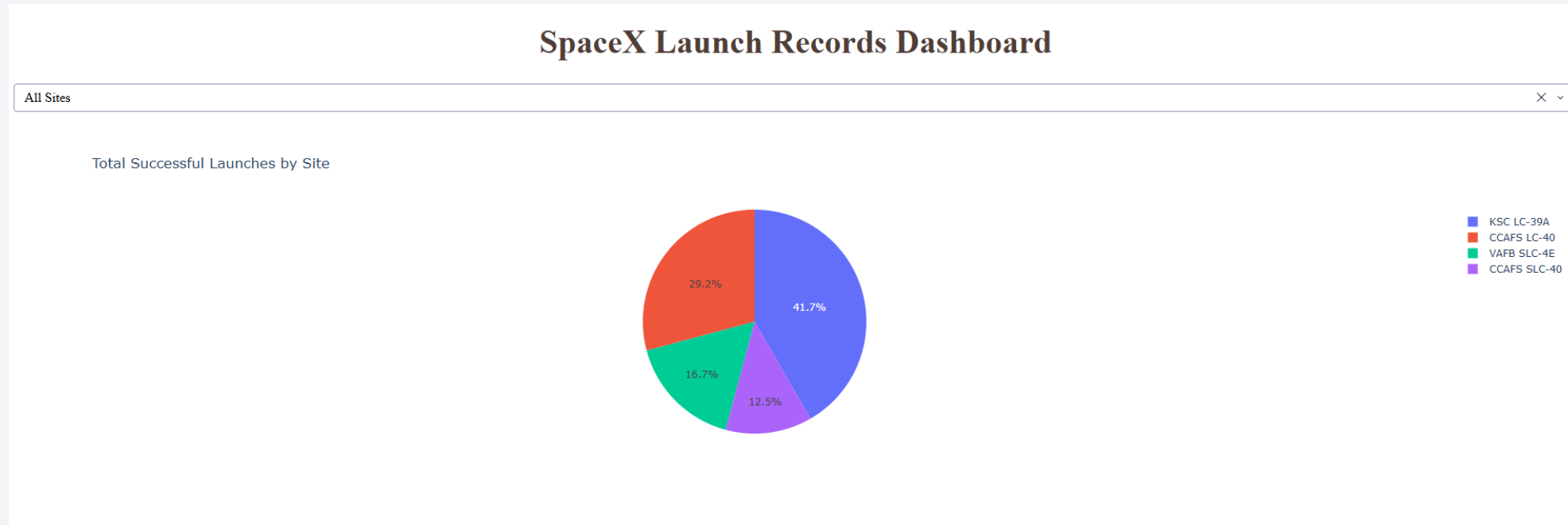




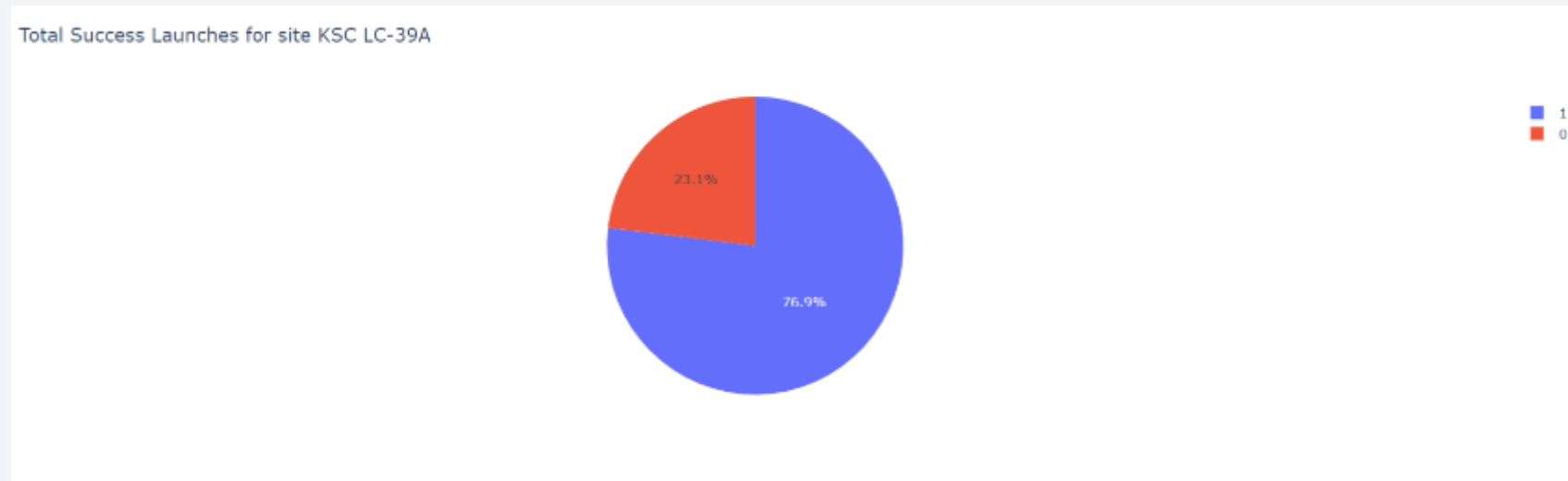
Section 4

Build a Dashboard with Plotly Dash

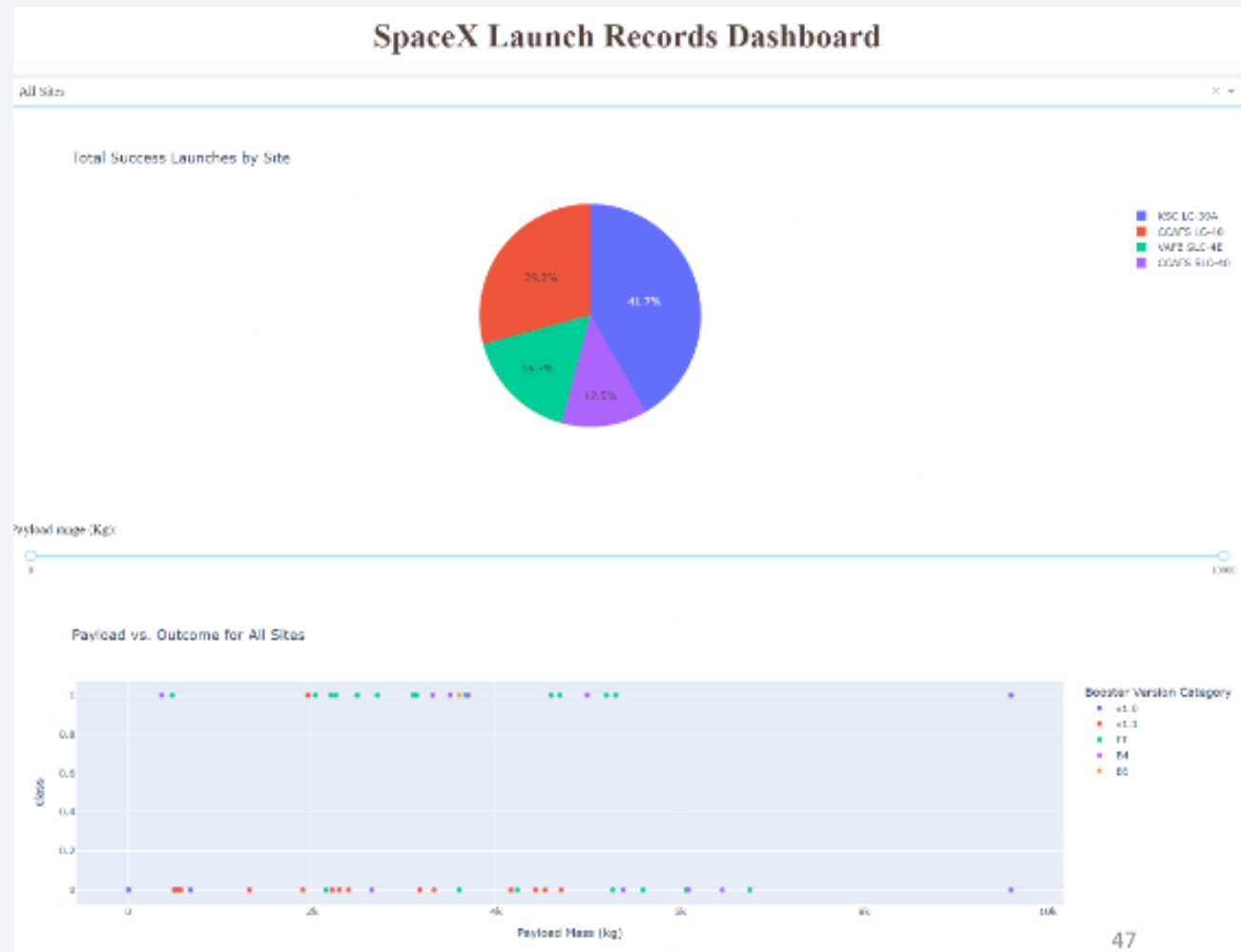
Plotly Dashboard Piechart for all sites



<Dashboard Screenshot 2>



Full Plotly Dashboard for all sites

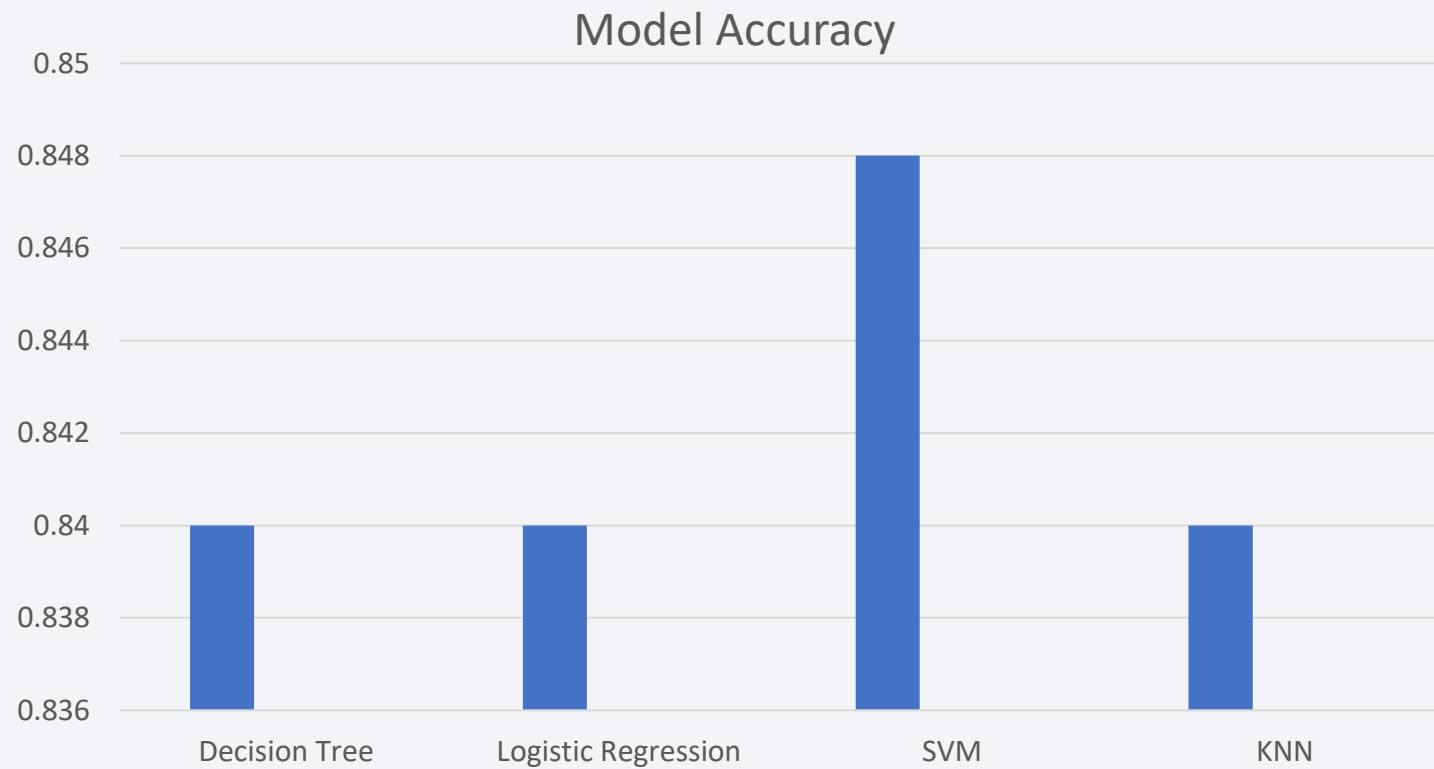




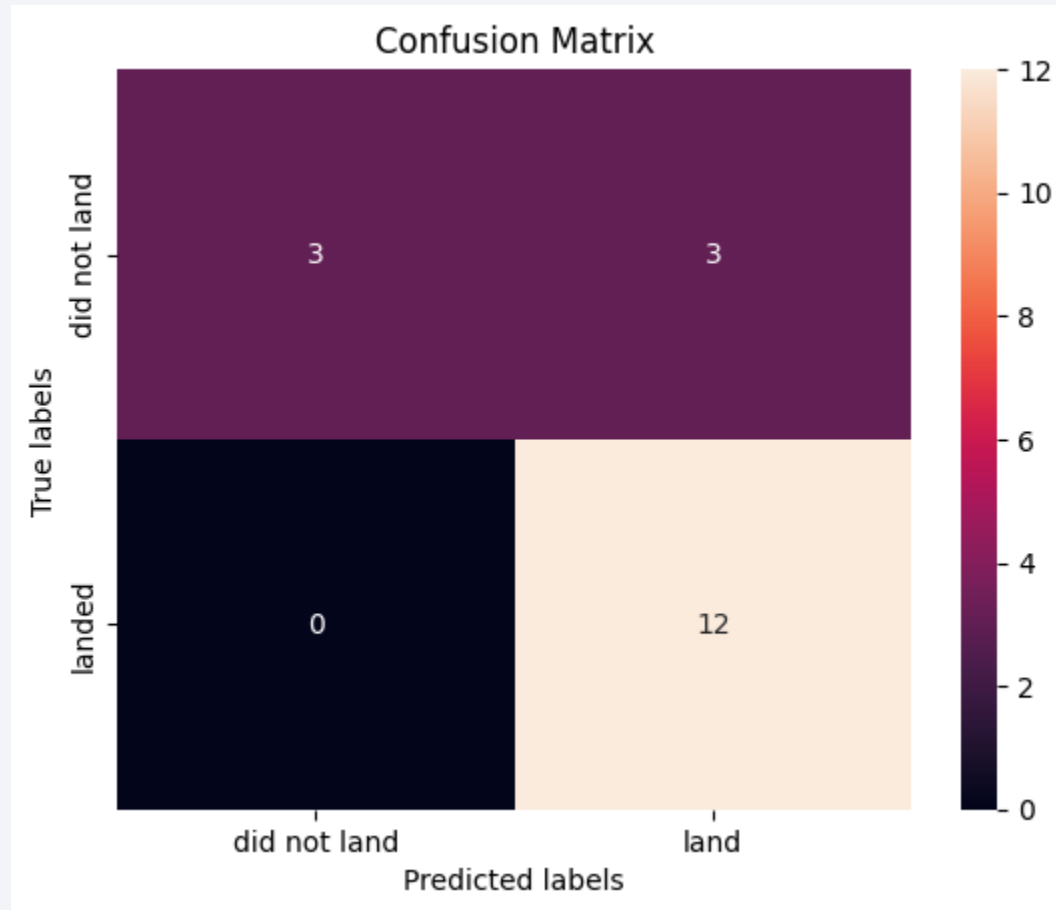
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



- Shows the confusion matrix of the best performing model

Conclusions

- After careful analysis it can be concluded that the launch site with the highest success rate out of the sites in the data set is “CCAFS LC-40” .
- The exploratory data analysis and interactive dashboard provides valuable insights to the trends in the data.
- Therefore, this case study provides a comprehensive analysis to the factors affecting the success of the landing of the first stage of Falcon 9 spaceships. The insights taken from this can help improve decision making when making a rocket making technology like SpaceX.

Thank you!

